Chapter 06

Hamilton Theory

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6.1 Hamiltonian of a dynamical system

In classical mechanics, there are three common approaches to solving problems. They are Newtonian mechanics, Lagrangian mechanics, and Hamiltonian mechanics.

Hamiltonian mechanics is a mathematically sophisticated formulation of classical mechanics. Hamiltonian mechanics was first formulated by ”William Rowan Hamilton” in 1833, starting from Lagrangian mechanics. Like Lagrangian mechanics, Hamiltonian mechanics is also equivalent to Newton’s laws of motion in the framework of classical mechanics.

Let L be the Lagrangian of the system with generalized coordinates $q_1, q_2, \dots, q_n$. The Lagrangian is a function of q_i , $\dot{q}_i$, t

$$L = L(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n, t)$$

$\frac{\partial L}{\partial \dot{q}_i}$ is called generalized momentum associated with q_i . It is denoted by p_i

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

The Hamiltonian of the system is defined by

$$H = \sum_{i=1}^{n} p_i \, \dot{q}_i - L \qquad (1)$$

$** \; H = H(p_i, q_i , t)$

By using the Lagrange's equation

$$\frac{d}{dx}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$$

We get $\dot{p}_i = \frac{\partial L}{\partial q_i}$

6.2 Hamiltonian equations

By the definition of H,

$$dH = d\left[\sum_{i=1}^{n} p_i\,\dot{q}_i\ - L\right]$$

$$= d\left[\sum_{i=1}^{n} p_i\,\dot{q}_i\right] - dL$$

$$= \sum_{i=1}^{n}(p_i\,d\dot{q}_i + \dot{q}_i dp_i) - \sum_{i=1}^{n}[\frac{\partial L}{\partial \dot{q}_i}dq_i + \frac{\partial L}{\partial \dot{q}_i}d\dot{q}_i]\ - \frac{\partial L}{\partial t}dt$$

$$= \sum_{i=1}^{n} p_i\,d\dot{q}_i + \sum_{i=1}^{n}\dot{q}_i dp_i - \sum_{i=1}^{n}\dot{p}_i\,dq_i - p_i d\dot{q}_i\ - \frac{\partial L}{\partial t}dt$$

$$= \sum_{i=1}^{n}(\dot{q}_i dp_i - \dot{p}_i dq_i)\ - \frac{\partial L}{\partial t}dt$$

$$dH = \sum_{i=1}^{n} (\dot{q}_i dp_i - \dot{p}_i dq_i) - \frac{\partial L}{\partial t} dt \qquad (2)$$

Using Lagrange's equations we can eliminate the variable $\dot{q}_i$ of the Hamiltonian H.

So, $H = H(p_i, q_i, t)$

$$dH = \sum_{i=1}^{n} [\frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial q_i} dq_i] + \frac{\partial H}{\partial t} dt \qquad (3)$$

By equating the corresponding coefficients of (2) and (3) equations, we get

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \qquad (4)$$

$$\frac{\partial H}{\partial q_i} = -\dot{p}_i \qquad (5)$$

and

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

(4) and (5) equations are called Hamilton's equations.

Suppose that H is independent of time t, then by (3) we can write,

$$dH = \sum_{i=1}^{n} [\frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial q_i} dq_i]$$

$$dH = \sum_{i=1}^{n} [\frac{\partial H}{\partial p_i}\frac{dp_i}{dt} + \frac{\partial H}{\partial q_i}\frac{dq_i}{dt}]$$

$$= \sum_{i=1}^{n} [\dot{q}_i \left(\frac{dp_i}{dt}\right) - \dot{p}_i \left(\frac{dq_i}{dt}\right)]$$

$$= \sum_{i=1}^{n} [\dot{q}_i \dot{p}_i - \dot{p}_i \dot{q}_i] = 0$$

Hence H is constant.

$$dH = \sum_{i=1}^{n} [\frac{\partial H}{\partial p_i} \frac{dp_i}{dt} + \frac{\partial H}{\partial q_i} \frac{dq_i}{dt}]$$

$$= \sum_{i=1}^{n} [\dot{q}_i \left(\frac{dp_i}{dt} \right) - \dot{p}_i \left(\frac{dq_i}{dt} \right)]$$

$$= \sum_{i=1}^{n} [\dot{q}_i \dot{p}_i - \dot{p}_i \dot{q}_i] = 0$$

Hence H is constant.

Kinetic energy of the system $T = \sum_{i=1}^{n} \frac{1}{2} m_i \dot{q}_i^2$ Then,

$$T(\lambda \dot{q}_i) = \sum_{i=1}^{n} \frac{1}{2} m_i \lambda^2 \dot{q}_i^2 = \lambda^2 \sum_{i=1}^{n} \frac{1}{2} m_i \dot{q}_i^2 = \lambda^2 T(\dot{q}_i)$$

Differentiate both sides w.r.t. λ,

$$\sum_{i=1}^{n} \frac{\partial T(\lambda \dot{q}_i)}{\partial (\lambda \dot{q}_i)} \frac{d(\lambda \dot{q}_i)}{d\lambda} = T(\dot{q}_i) \ \frac{d\lambda^2}{d\lambda}$$

$$\sum_{i=1}^{n} \frac{\partial T(\lambda \dot{q}_i)}{\partial (\lambda \dot{q}_i)} \dot{q}_i = T(\dot{q}_i) \ 2\lambda$$

set λ = 1,

$$\sum_{i=1}^{n} \frac{\partial T(\lambda \dot{q}_i)}{\partial(\dot{q}_i)} \dot{q}_i = T(\dot{q}_i)\ 2$$

Simply we can write,

$$\sum_{i=1}^{n} \frac{\partial T}{\partial(\dot{q}_i)} \dot{q}_i = 2T$$

Since $p_i = \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial(T-V)}{\partial \dot{q}_i} = \frac{\partial T}{\partial \dot{q}_i}$Then,

$$\sum_{i=1}^{n} p_i\, \dot{q}_i = 2T$$

So,

$$H = \sum_{i=1}^{n} p_i \, \dot{q}_i - L = 2T - (T - V)$$

$$= T + V = E(\text{Total energy of the system}) \qquad (6)$$

Example 01

Consider the following Lagrangian,

$$L = A\dot{x}^2 + B\dot{y} + C\dot{x}\dot{y} + \frac{D}{\sqrt{x^2 + y^2}}$$

Where A,B,C,D are constants and C 6= 0. Determine the Hamiltonian.

Solution:

By equation (1)

$$H = \sum_{i=1}^{n} p_i \, \dot{q}_i - L, \qquad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Generalized coordinates are x and y,

$$p_x = \frac{\partial L}{\partial \dot{x}} = 2A\dot{x} + \mathrm{C}\dot{y} \qquad (7)$$

$$p_y = \frac{\partial L}{\partial \dot{y}} = \mathrm{B} + \mathrm{C}\dot{x} \qquad (8)$$

By equation (7) and (8) $\dot{x} = \frac{p_y - B}{C}$ $\dot{y} = \frac{p_x}{C} - 2A\left(\frac{p_y - B}{C^2}\right)$

$$H = p_x\dot{x} + p_y\dot{y} - L$$

$$H = p_x\left[\frac{p_y - B}{C}\right] + p_y\left[\frac{p_x}{C} - 2A\left(\frac{p_y - B}{C^2}\right)\right] - A\left[\frac{p_y - B}{C}\right] - B\left[\frac{p_x}{C} - 2A\left(\frac{p_y - B}{C^2}\right)\right]$$

$$\frac{p_y - B}{C}\left[\frac{p_x}{C} - 2A\left(\frac{p_y - B}{C^2}\right)\right] - \frac{D}{\sqrt{x^2 + y^2}}$$

Example 02

A particle moves in the O_{xy} plane under the influence of the conservative force whose potential function is V, a function of r only.

1. Find the Hamiltonian of the system.
2. Write down Hamilton's equation.

Solution:

1. Potential energy = V(r)

Kinetic energy,

$$T = \frac{1}{2}m[\dot{r}^2 + \mathrm{r}^2\dot{\theta}^2]$$

$$L = T - \mathrm{V}$$
$$= \frac{1}{2}m[\dot{r}^2 + \mathrm{r}^2\dot{\theta}^2] - \mathrm{V(r)}$$

Method (I)

$$H = T + V$$

$$= \frac{1}{2} m[\dot{r}^2 + \mathrm{r}^2 \dot{\theta}^2] + \mathrm{V(r)}$$

$H = H(q_i\ , p_i, t\ )$

Generalized coordinates are r and θ

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

$$p_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \quad \Rightarrow \dot{r} = \frac{p_r}{m}$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \mathrm{mr}^2 \dot{\theta} \ \Rightarrow \ \dot{\theta} = \frac{p_\theta}{mr^2}$$

$$H = \frac{1}{2} m \left[\left(\frac{p_r}{m} \right)^2 + \mathrm{r}^2 \left(\frac{p_\theta}{mr^2} \right)^2 \right] + \mathrm{V(r)}$$

Method (II)

$$H = \sum_{i=1}^{n} p_i \, \dot{q}_i - L$$

$$= p_r \dot{r} + p_\theta \dot{\theta} - \mathrm{L}$$

$$= p_r \left(\frac{p_r}{m}\right) + p_\theta \left(\frac{p_\theta}{mr^2}\right) - \frac{1}{2} m[\dot{r}^2 + \mathrm{r}^2 \dot{\theta}^2] + \mathrm{V(r)}$$

$$= \frac{1}{2} m \left[\left(\frac{p_r}{m}\right)^2 + \mathrm{r}^2 \left(\frac{p_\theta}{mr^2}\right)^2\right] + \mathrm{V(r)}$$

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \qquad\qquad \frac{\partial H}{\partial q_i} = -\dot{p}_i$$

$$\dot{r} = \frac{p_r}{m} \tag{9}$$

$$\dot{\theta} = \frac{p_\theta}{mr^2} \tag{10}$$

$$-\dot{p}_r = \frac{\partial H}{\partial r} = -\frac{1}{m}\frac{p_\theta^2}{\mathrm{r}^3} + \frac{d}{dr}V(r)$$

$$\dot{p}_r = \frac{1}{m}\frac{p_\theta^2}{\mathrm{r}^3} - \frac{d}{dr}V(r) \tag{11}$$

$$\dot{p}_\theta = 0 \tag{12}$$

(9), (10), (11) and (12) are Hamilton's equations.

6.3 Canonical transformation and Generating functions

For a given system of particles, we can define different sets of generalized coordinates. In this section we begin by considering canonical transformations. These are transformations that preserve the form of Hamilton's equations. A transformation between two sets of generalized coordinates is called a "Canonical Transformation", if it leaves the Hamilton's equations invariant.

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Hamilton's Equation

The new coordinates P_i and Q_i can also be used to describe the Hamiltonian of the system.

A canonical transformation is one that takes us from the p_i, q_i to a set of coordinates

P_i and Q_i such that

$$\dot{Q}_i = \frac{\partial H}{\partial P_i}$$

$$-\dot{P}_i = \frac{\partial H}{\partial Q_i}$$

$K = K(P_i\,, Q_i, t\,)$

Theorem 1. *The transformation* $q_i\,, p_i \implies Q_i\,, P_i$ *is canonical if* $\sum(p_i dq_i - P_i dQ_i)$ *is an exact differential.*

Example 1

Show that the following transformation of one degree of freedom is canonical.

$$P = \frac{1}{2}(p^2 + q^2)$$
$$Q = \tan^{-1}\left(\frac{q}{p}\right)$$

Solution: Method I

Since $Q = \tan^{-1}\left(\frac{q}{p}\right)$

$$dQ = \frac{1}{1+\left(\frac{q}{p}\right)^2}\frac{pdq-qdp}{p^2} = \frac{pdq-qdp}{p^2+q^2}$$

$$\sum(p_i dq_i - P_i dQ_i) = pdq - PdQ$$

$$= pdq - \frac{1}{2}(p^2+q^2)\frac{pdq-qdp}{p^2+q^2}$$

$$= pdq - \frac{1}{2}pdq + \frac{1}{2}qdp$$

$$= \frac{1}{2}pdq + \frac{1}{2}qdp$$

$$= \frac{1}{2}(pdq + qdp) = d(\frac{1}{2}pq)$$

So the transformation is canonical.

Solution: Method II

Jacobian of the transformation is

$$M = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix}$$

If $MJM^T = J$ *then the transformation is canonical. Where* $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

$$\frac{\partial Q}{\partial q} = \frac{1}{p\left(1 + \left(\frac{q}{p}\right)^2\right)} = \frac{p}{p^2 + q^2}$$

$$\frac{\partial Q}{\partial p} = -\frac{q}{p^2\left(1 + \left(\frac{q}{p}\right)^2\right)} = \frac{-q}{p^2 + q^2}$$

$$\frac{\partial P}{\partial q} = q$$

$$\frac{\partial P}{\partial p} = p$$

$$M = \begin{pmatrix} \frac{p}{p^2+q^2} & \frac{-q}{p^2+q^2} \\ q & p \end{pmatrix}$$

$$MJM^T = \begin{pmatrix} \frac{p}{p^2+q^2} & \frac{-q}{p^2+q^2} \\ -p & q \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{p}{p^2+q^2} & q \\ \frac{-q}{p^2+q^2} & p \end{pmatrix}$$

$$= \begin{pmatrix} \frac{q}{p^2+q^2} & \frac{p}{p^2+q^2} \\ -p & q \end{pmatrix} \begin{pmatrix} \frac{p}{p^2+q^2} & q \\ \frac{-q}{p^2+q^2} & p \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J$$

So the transformation is canonical.

Example 02

Show that the transformation

$$P = \ln(sinp) \qquad Q = qtanp$$

is canonical.

Solution: Method I

$$Q = qtan(p)$$

$$dQ = d(qtan(p))$$

$$pdq - PdQ = pdq - \ln(sinp)\,[\tan(p)\,dq + qsec^2(p)dp]$$

$$= (p - \ln(sinp))\tan(p))dq + q\ln(sinp)\,sec^2(p)dp$$

Since,

$$d(pq - qln(\sin(p))\tan(p)) = d[q(p - \ln(\sin(p))\tan(p))]$$

$$= (p - \ln(\sin(p))\tan(p))dq - qln(sinp)\sec^2(p)dp$$

Therefore,

$$pdq - P\,dQ = d(pq - qln(\sin(p))\tan(p))$$

So the transformation is canonical.

Solution: Method II

Jacobian of the transformation is

$$M = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} = \begin{pmatrix} \tan(p) & qsec^2(p) \\ 0 & \frac{1}{\tan(p)} \end{pmatrix}$$

$$MJM^T = \begin{pmatrix} \tan(p) & qsec^2(p) \\ 0 & \frac{1}{\tan(p)} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \tan(p) & 0 \\ qsec^2(p) & \frac{1}{\tan(p)} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J$$

So the transformation is canonical. Where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

Generating functions

A transformation $p_i, q_i \longrightarrow P_i$ and Q_i is canonical, provided that there is a function G called the generating function, satisfying

$$\frac{d}{dt}G = L(q_i\,, \dot{q}_i, t\,) - L\big(Q_i\,, \dot{Q}_i, t\,\big) \qquad (1)$$

here L's are the Lagrangians with respect to the old and new variables.

We can express the generating function G using one of the two old variables and one of the two new variables.

	q_i	p_i
Q_i	F_1	F_2
P_i	F_3	F_4

Result (I):

When the generating function G is expresses using q_i, Q_i and t it is denoted by F_1 and we get

$$p_i = \frac{\partial F_1}{\partial q_i} \qquad P_i = -\frac{\partial F_1}{\partial Q_i} \qquad K = H + \frac{\partial F_1}{\partial t}$$

Proof. $F_1 = F_1(q_i\,, Q_i, t\,)$

$$dF_1 = \sum_i \left[\frac{\partial F_1}{\partial q_i} dq_i + \frac{\partial F_1}{\partial Q_i} dQ_i\right] + \frac{\partial F_1}{\partial t} dt \qquad (1)$$

Further,

$$\frac{dG}{dt} = L(q_i\,, \dot{q}_i, t\,) - L\big(Q_i\,, \dot{Q}_i, t\,\big)$$

$$= \sum p_i \dot{q}_i - H - \sum P_i \dot{Q}_i + K$$

$$dF_1 = \sum p_i \dot{q}_i \, dt - H dt - \sum P_i \dot{Q}_i dt + K \, dt$$

$$dF_1 = \sum p_i dq_i - H dt - \sum P_i dQ_i + K \, dt \qquad (2)$$

by (1) and (2)

$$p_i = \frac{\partial F_1}{\partial q_i} \qquad p_i = -\frac{\partial F_1}{\partial Q_i} \qquad K = H + \frac{\partial F_1}{\partial t}$$

Result (II):

When the generating function G is expressed using p_i, Q_i and t it is denoted by F_2 and we get

$$q_i = -\frac{\partial F_2}{\partial p_i} \qquad P_i = -\frac{\partial F_2}{\partial Q_i} \qquad K = H + \frac{\partial F_2}{\partial t}$$

Proof. $F_2 = F_2(p_i\,, Q_i, t\,)$

$$dF_2 = \sum_i \left[\frac{\partial F_2}{\partial p_i} dp_i + \frac{\partial F_2}{\partial Q_i} dQ_i\right] + \frac{\partial F_2}{\partial t} dt \qquad (3)$$

Further,

$$\frac{dG}{dt} = L(q_i\,, \dot{q}_i, t\,) - L\big(Q_i\,, \dot{Q}_i, t\,\big)$$

$$= \sum p_i \dot{q}_i - H - \sum P_i \dot{Q}_i + K$$

$$= \sum p_i \dot{q}_i - \sum P_i \dot{Q}_i + (K - H)$$

$$= \sum p_i \dot{q}_i + \sum \dot{p}_i q_i - \sum \dot{p}_i q_i - \sum P_i \dot{Q}_i + (K - H)$$

$$\frac{dG}{dt} - \sum [p_i \dot{q}_i + \dot{p}_i q_i] = -(\sum \dot{p}_i q_i + \sum P_i \dot{Q}_i) + (K - H)$$

$$\frac{d}{dt}[G - \sum (p_i q_i)] = -(\sum \dot{p}_i q_i + \sum P_i \dot{Q}_i) + (K - H)$$

$$dF_2 = \sum (-q_i dp_i - P_i dQ_i)) + (K - H)dt$$

$$q_i = -\frac{\partial F_2}{\partial p_i} \qquad P_i = -\frac{\partial F_2}{\partial Q_i} \qquad K = H + \frac{\partial F_2}{\partial t}$$

Result (III):

When the generating function G is expresses using q_i, P_i and t it is denoted by F_3 and we get

$$p_i = \frac{\partial F_3}{\partial p_i} \qquad P_i = -\frac{\partial F_3}{\partial P_i} \qquad K = H + \frac{\partial F_3}{\partial t}$$

Proof. $F_3 = F_3(q_i\,, P_i, t\,)$

$$dF_3 = \sum_i \left[\frac{\partial F_3}{\partial q_i} dq_i + \frac{\partial F_3}{\partial P_i} dP_i\right] + \frac{\partial F_3}{\partial t} dt \qquad (5)$$

Further,

$$\frac{dG}{dt} = L(q_i\,, \dot{q}_i, t\,) = L\big(Q_i\,, \dot{Q}_i, t\,\big)$$

$$= \sum p_i \dot{q}_i - H - \sum P_i \dot{Q}_i + K$$

$$= \sum p_i \dot{q}_i + \sum Q_i \dot{P}_i - \sum Q_i \dot{P}_i \sum P_i \dot{Q}_i - H + K$$

$$\frac{dG}{dt} = \sum [p_i \dot{q}_i + \dot{p}_i Q_i] - \sum (Q_i \dot{P}_i + P_i \dot{Q}_i) + K - H \qquad (13)$$

$$\frac{d}{dt}[G + \sum (P_i Q_i)] = \sum [p_i \dot{q}_i + \dot{P}_i Q_i] + K - H$$

$$d[G + \sum (P_i Q_i)] = \sum [p_i dq_i + Q_i dP_i] + K - H$$

$$dF_3 = \sum [p_i dq_i + Q_i dP_i] + K - H$$

$$q_i = \frac{\partial F_3}{\partial q_i} \qquad P_i = \frac{\partial F_3}{\partial P_i} \qquad K = H + \frac{\partial F_3}{\partial t}$$

Result (IV):

When the generating function G is expresses using p_i, P_i and t it is denoted by F_4 and we get

$$q_i = -\frac{\partial F_4}{\partial p_i} \qquad Q_i = \frac{\partial F_4}{\partial P_i} \qquad K = H + \frac{\partial F_4}{\partial t}$$

Proof. $F_4 = F_4(p_i\ , P_i, t\ )$

$$dF_4 = \sum_i \left[\frac{\partial F_4}{\partial p_i} dp_i + \frac{\partial F_4}{\partial P_i} dP_i\right] + \frac{\partial F_4}{\partial t} dt \qquad (7)$$

Further,

$$\frac{dG}{dt} = L(q_i\ , \dot{q}_i, t\ ) = L\big(Q_i\ , \dot{Q}_i, t\ \big)$$

$$= \sum p_i \dot{q}_i - H - \sum P_i \dot{Q}_i + K \qquad (14)$$

$$dF_4 = \sum [-q_i dp_i + Q_i dP_i] + (K - H)dt$$

$$q_i = -\frac{\partial F_4}{\partial p_i} \qquad Q_i = \frac{\partial F_4}{\partial P_i} \qquad K = H + \frac{\partial F_4}{\partial t}$$

Summary of the different types of Generators

There are four type of generating functions to consider and these are,

Generators	Derivatives	Relationship
$F = F_1(q_i, Q_i, t)$	$p_i = \frac{\partial F_1}{\partial q_i} \quad P_i = -\frac{\partial F_1}{\partial Q_i}$	$K = H + \frac{\partial F_1}{\partial t}$
$F = F_2(p_i, Q_i, t)$	$q_i = -\frac{\partial F_2}{\partial p_i} \quad P_i = -\frac{\partial F_2}{\partial Q_i}$	$K = H + \frac{\partial F_2}{\partial t}$
$F = F_3(q_i, P_i, t)$	$p_i = \frac{\partial F_3}{\partial q_i} \quad Q_i = \frac{\partial F_3}{\partial P_i}$	$K = H + \frac{\partial F_3}{\partial t}$
$F = F_4(p_i, P_i, t)$	$q_i = -\frac{\partial F_4}{\partial p_i} \quad Q_i = \frac{\partial F_4}{\partial P_i}$	$K = H + \frac{\partial F_4}{\partial t}$

Example 01:

Show the following transformation is canonical and find the generating function.

$$P = q \qquad Q = -p$$

Solution

$$\frac{\partial F_1}{\partial q_i} = p = -Q$$

By integrating partially with respect to q, we get

$$F_1 = -Qq + f(Q)$$

$$\frac{\partial F_1}{\partial Q} = -P = -q$$

$$-q + \frac{df(Q)}{dQ} = -q \Rightarrow \frac{df(Q)}{dQ} = 0 \Rightarrow f(Q) = c, constant.$$

$$F_1 = -Qq + c$$

So, the generating function $F_1 = -Qq$ *, neglecting the constant.*

Example 02:

For the canonical transformation,

$$P = \ln(\sin(p)) \qquad Q = qtan(p)$$

Find the generating function in terms of P and q.

Solution

$$\text{p}= \sin^{-1}(e^P) \qquad Q = qtan(p)$$

$$p = \frac{\partial F_3}{\partial q} \Rightarrow \frac{\partial F_3}{\partial q} = \sin^{-1}(e^P)$$

$$F_3 = q\sin^{-1}(e^P) + f(P,t)$$

$$Q = \frac{\partial F_3}{\partial P}$$

$$= q\frac{e^p}{\sqrt{1-e^{2p}}} + \frac{\partial f(P,t)}{\partial P} = q\tan(\sin^{-1}(e^p))$$

$$= q\frac{e^p}{\sqrt{1-e^{2p}}} + \frac{\partial f(P,t)}{\partial P} = q\frac{e^p}{\sqrt{1-e^{2p}}}$$

$$\Rightarrow \frac{\partial f(P,t)}{\partial P} = 0 \Rightarrow f(P,t) = f(t)$$

So the generating function is

$$F_3 = q\sin^{-1}(e^p) + f(t)$$

Example 03:

Show that the transformation for a system of one degree of freedom,

$$Q = qcos(\alpha) - psin(\alpha) \qquad P = qsin(\alpha) + pcos(\alpha)$$

Find the generating function for the transformation.

Solution:

$$M = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} = \begin{pmatrix} cos(\alpha) & -sin(\alpha) \\ sin(\alpha) & cos(\alpha) \end{pmatrix}$$

$$MJM^T = \begin{pmatrix} cos(\alpha) & -sin(\alpha) \\ sin(\alpha) & cos(\alpha) \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} cos(\alpha) & sin(\alpha) \\ -sin(\alpha) & cos(\alpha) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J$$

Let us try to find a generating function of the first type, $F_1(q, Q, t)$ *for the transformation.*

$$\text{p} = \frac{\partial F_1}{\partial q} \qquad P = -\frac{\partial F_1}{\partial Q}$$

$$Q = qcos(\alpha) - psin(\alpha) \Rightarrow p = qcot(\alpha) - Qcsc(\alpha)$$

$$P = qsin(\alpha) + pcos(\alpha) = q(\frac{\cos^2(\alpha)}{\sin(\alpha)} + \sin(\alpha)) - Q\,cot(\alpha) = qcsc(\alpha) - Qcot(\alpha)$$

$$\frac{\partial F_1}{\partial q} = qcot(\alpha) - Qcsc(\alpha)$$

$$F_1 = \frac{q^2}{2} cot(\alpha) - Qqcsc(\alpha) + f(Q)$$

$$\frac{\partial F_1}{\partial Q} = -qcsc(\alpha) + f'(Q)$$

$$= -qsin(\alpha) + pcos(\alpha)$$

$$= -qsin(\alpha) + Qcot(\alpha)$$

$$f'(Q) = Qcot(\alpha)$$

$$f(Q) = \frac{Q^2}{2} cot(\alpha)$$

Generating function is,

$$F_1(q, Q, t) = \frac{1}{2}(q^2 + Q^2)\cot(\alpha) - Qpcsc(\alpha)$$

6.4 Hamiltonian-Jacobi Equation

A dynamical problem is solved when we determine relationships for the generalized coordinates and momenta in terms of the initial conditions and the time. The initial conditions can be expresses $as\ q_0\ and\ p_0$ by which we mean the constant values of the

coordinates (q_i) and the momenta (p_i) at time t=0.

The solution of the mechanical problem is the set of relationships

$$q = q(q_0\ , p_0\ , t) \qquad (1)$$

$$\mathrm{p} = p(q_0\ , p_0\ , t)$$

These equations can be considered to be a set of transformation equations between the canonical variables q, p and a new set $as\ q_0, p_0$

Finding a solution to the mechanical problem is equivalent to finding the transformation from q, p to q_0, p_0. But q_0 and p_0 are constants, therefore we need to find a transformation from the canonical variables (q, p) to a set of variables which are constants.

A canonical transformation takes us from the Hamiltonian $H = H(p, q, t)$ to the transformed Hamiltonian $K = K(P, Q, t)$, where

$$\dot{Q}_i = \frac{\partial K}{\partial P_i} \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i} \qquad K = H + \frac{\partial F}{\partial t}$$

If the new variables are constants then we have,

$$\dot{Q}_i = 0 \qquad \dot{P}_i = 0$$

(We want the new variables to be the constant initial values of the coordinates and momenta)

The requirement that the new variables be constant is guaranteed quite simply by requiring that K be a constant. In fact, if we set $K = 0$, then obviously $\dot{Q}_i = 0 \; and \; \dot{P}_i = 0.$

Setting $K = 0$ yields

$$H + \frac{\partial F}{\partial t_1} = 0$$

Here F is the generating function for the canonical transformation.

We shall choose F to have the form $F = F_3(q, P, t)$. This is a reasonable choice because this generating function takes us from $p\ to\ P\ and\ from\ q\ to\ Q$. We previously denoted this generating function by F_3, but now, for historical reasons, we shall denote it by S.

$$S = S(q, P, t)$$

In Hamilton-Jacobi theory, the function S is called Hamilton's principal function.

From previous section,

$$p_i = \frac{\partial S}{\partial q_i} \qquad Q_i = \frac{\partial S}{\partial P} \qquad K = H + \frac{\partial S}{\partial t}$$

If we can determine S, then we will be able to generate the transformation equations (1) and problem is solved.

Then,

$$H(q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n, ; t) + \frac{\partial S}{\partial t} = 0$$

but $p_i = \frac{\partial S}{\partial q_0}$ so we can write

$$H\left(q_1, \dots, q_n, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}; t\right) + \frac{\partial S}{\partial t} = 0$$

This is a partial differential equation for S.

It is called the **Hamilton-Jacobi equation**.

Recall that the new variables $P\ and\ Q$ are the constant initial conditions. To emphasize this, one usually write $\alpha_i\ for\ P_i\ and\ \beta_i\ for\ Q_i$. Then,

$$S = S(q_i, \alpha_i, t)$$

$$p_i = \frac{\partial S}{\partial q_i} = \frac{\partial S(q_i, \alpha_i, t)}{\partial q_i}$$

$$\beta_i = Q_i = \frac{\partial S}{\partial \alpha_i} = \frac{\partial S(q_i, \alpha_i, t)}{\partial \alpha_i}$$

$$q_i = q_i(\alpha_i, \beta_i, t) \qquad (2)$$

$$p_i = \frac{\partial}{\partial q_i} \mathrm{S}(\mathrm{q_i}(\alpha_i, \beta_i, t), \alpha_i, \mathrm{t})$$

$$p_i = p_i(\alpha_i, \beta_i, t) \qquad (3)$$

(2) and (3) are the desired solution to the dynamical problem.

Example 01

State the Hamilton-Jacobi equation.

The Hamiltonian of a dynamical system with the generalized coordinate q and generalized momentum p, is given by

$$H = \frac{1}{2a}p^2 + 2aq^2$$

where a is a non-zero constant. Write down the Hamilton-Jacobi equation for the system.

a) Explaining all the necessary steps, show that the generating function of the corresponding transformation can be expresses as

$$S(q, P, t) = -Pt + \sqrt{2a} \int \sqrt{P - 2aq^2} dq$$

b) If initially $q = 0$, derive that $q = csin(t + n\pi)$, where c is a constant and n is an integer.

Solution

Hamilton-Jacobi equation,

$$H\left(q_1, \dots, q_n, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}; t\right) + \frac{\partial S}{\partial t} = 0$$

(a) $H = \frac{1}{2a}p^2 + 2aq^2$ where a is a non-zero constant From Hamilton-Jacobi equation,

$$-\frac{\partial S}{\partial t} = H\left(q, \frac{\partial S}{\partial q}, t\right)$$

$$\frac{\partial S}{\partial q} = p$$

Therefore,

$$-\frac{\partial S}{\partial t} = \frac{1}{2a}\left(\frac{\partial S}{\partial q}\right)^2 + 2\mathrm{aq}^2$$

$$S(\alpha, q, t) = W(\alpha, q) + \mathrm{V}(\alpha, t), \alpha \; is \; a \; constant$$

$$-\frac{\partial V}{\partial q} = \frac{1}{2a}\left(\frac{\partial W}{\partial q}\right)^2 + 2\mathrm{aq}^2$$

The left hand-side depends only on t and the right hand-side depends only on q Since q and t are independent, both sides must be equal to the same constant. Let us denote this constant by α, then

$$-\frac{\partial V}{\partial q} = \alpha \;\Rightarrow\; V = -\alpha t$$

and

$$\frac{1}{2a}\left[\left(\frac{\partial W}{\partial q}\right)^2 + 4\mathrm{a}^2\mathrm{q}^2\right] = \alpha$$

$$W = \int \sqrt{2a\alpha - 4\mathrm{a}^2\mathrm{q}^2}\,dq$$

Therefore,

$$S(\alpha, q, t) = -\alpha t + \sqrt{2a}\int \sqrt{\alpha - 2aq^2}\,dq$$

Note that α = P

$$S(P, q, t) = -Pt + \sqrt{2a}\int \sqrt{P - 2aq^2}\,dq$$

(b)

$$\beta = \frac{\partial S}{\partial P} = -t + \sqrt{2a}\int \frac{1}{2\sqrt{P\ - 2aq^2}}dq$$

$$\beta = -t + \frac{1}{2}\sin^{-1}(\sqrt{\frac{2a}{p}}q)$$

$$\sqrt{\frac{P}{2a}}\ \sin\big(2(\beta + t)\big) = q$$

When $t = 0\ \ q = 0\ \ \Longrightarrow\ 2\beta\ =\ n\pi, n\ \in\ \mathbb{Z}$
$q = csin(2t + n\pi)$.

Example 02:

The Hamiltonian of a one dimensional harmonic oscillator is given by

$$H = \frac{1}{2m}p^2 + \frac{1}{2}kx^2$$

where m,p,x and k are the mass of the oscillator, generalized momentum, generalized coordinate and a constant respectively.

(a) Write down the Hamilton-Jacobi equation for the oscillator.

(b) Explaining the necessary steps, show that the generating function of the corresponding transformation can be expressed as,

$$S(x,t) = -\beta t + \int \sqrt{m\,(2\beta - kx^2)}dx$$

where β is a constant.

(c) If initially x=0, derive that

$$x(t) = \sqrt{\frac{2\beta}{k}}\sin\left(n\pi + \sqrt{\frac{k}{m}}t\right) \quad where\ n\ is\ an\ integer$$

Solution:

Hamilton-Jacobi equation,

$$H\left(q_1, \dots, q_n, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}; t\right) + \frac{\partial S}{\partial t} = 0$$

Hamilton-Jacobi equation for the oscillator is

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0$$

(b) $S(q, \mathrm{P}, t)=W(\mathrm{P},q)+\mathrm{V}(\mathrm{P}, t)$,

$\mathrm{P}=\alpha,\ \alpha\ is\ a\ constant.$

$S(q, \alpha, t)=W(\alpha,q)+\mathrm{V}(\alpha, t)$,

$$\frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2+\frac{1}{2}kq^2=-\frac{\partial S}{\partial t}$$

$$\frac{1}{2m}\left(\frac{\partial W}{\partial q}\right)^2+\frac{1}{2}kq^2=-\frac{\partial V}{\partial t}$$

Since q and t are independent variables

$$-\frac{\partial V}{\partial t} = \alpha \;\; and \;\; \frac{1}{2m}\left(\frac{\partial W}{\partial q}\right)^2 + \frac{1}{2}kq^2 = \alpha$$

$$V = -\alpha t \qquad W = \int \sqrt{2\alpha m - mk\mathrm{q}^2}dq$$

$$S(q,t) = -\alpha t + \int \sqrt{m(2\alpha - k\mathrm{q}^2)}dx$$

Note that q = x and β= α

$$S(P,q,t) = -\beta t + \sqrt{2a}\int \sqrt{m(2\beta - k\mathrm{q}^2)}dx$$

$$\beta = -t + \sqrt{m}\int \frac{1}{\sqrt{2P\ - kx^2}}dx$$

$$(\beta + t) = \sqrt{\frac{m}{k}}\ \sin^{-1}\sqrt{\frac{kx^2}{2P}}$$

$$\mathrm{Sin}(\sqrt{\frac{k}{m}}\ (\beta + t)) = \sqrt{\frac{k}{2P}}x$$

$$\sqrt{\frac{2P}{k}}\mathrm{Sin}(\sqrt{\frac{k}{m}}\ (\beta + t)) = x$$

$$t = 0, x = 0\sqrt{\frac{k}{m}}\beta\ =\ n\pi\ \ Therefore,$$

$$\sqrt{\frac{2P}{k}}\mathrm{Sin}(n\pi + \sqrt{\frac{k}{m}}\ \mathrm{t})\ = x$$

Example 03:

State Hamilton-Jacobi equation

The Hamiltonian of a dynamical system with the generalized coordinate q and generalized momentum p is given by,

$$H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

where m and ω are non-zero constants.

Write down Hamilton-Jacobi equation for the system.

(I) Explaining all the necessary steps, show that the generating function of the corresponding transformation can be expressed as,

$$S(q, Q, t) = -Pt + \int \sqrt{m\,(2P - m\omega^2 q^2)}\, dq$$

(II) If initially $q = 0$, derive that $q = csin(n\pi + \omega t)$ where c is a constant and n is an integer.

Solution:

Hamilton-Jacobi equation,

$$H\left(q_1, \dots, q_n, \frac{\partial S}{\partial q_1}, \dots, \frac{\partial S}{\partial q_n}; t\right) + \frac{\partial S}{\partial t} = 0$$

Hamilton-Jacobi equation for the system is

$$H\left(q, \frac{\partial S}{\partial q}, t\right) + \frac{\partial S}{\partial t} = 0$$

$$S(q, P, t) = W(P, q) + \mathrm{V}(P, t)$$

$$-\frac{\partial S}{\partial t} = \frac{1}{2m} p^2 + \frac{\mathrm{m}\omega^2}{2} \mathrm{q}^2$$

$$-\frac{\partial S}{\partial t} = \frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2 + \frac{\mathrm{m}\omega^2}{2}\mathrm{q}^2$$

Then,

$$-\frac{\partial V}{\partial t} = \frac{1}{2m}\left(\frac{\partial W}{\partial q}\right)^2 + \frac{\mathrm{m}\omega^2}{2}\mathrm{q}^2$$

Since q and t are independent,

$$-\frac{\partial V}{\partial t} = \alpha \;\; and \;\; \frac{1}{2m}\left(\frac{\partial W}{\partial q}\right)^2 + \frac{\mathrm{m}\omega^2}{2}\mathrm{q}^2 = \alpha$$

$$V = -\alpha t \qquad W = \int \sqrt{2m\alpha - m^2\omega^2\mathrm{q}^2}dq = \int \sqrt{m(2\alpha - m\omega^2\mathrm{q}^2)}dq$$

Note that P = α

$$S(q,P,t) = -Pt + \int \sqrt{m(2P - m\omega^2 q^2)}dq$$

(II)

$$Q = \frac{\partial S}{\partial P} = \beta$$

$$\beta = -t + \sqrt{m}\int \frac{1}{\sqrt{2P - m\omega^2 q^2}}dq = -t + \frac{1}{\omega}\sin^{-1}\sqrt{\frac{m\omega^2 q^2}{2P}}$$

$$\omega\beta = -\omega t + \sin^{-1}\sqrt{\frac{m\omega^2 q^2}{2P}}$$

$$\mathrm{Sin}(\omega\beta + \omega t)) = \sqrt{\frac{m\omega^2}{2P}}q$$

$$q = \sqrt{\frac{2P}{m\omega^2}}\,\mathrm{Sin}(\omega\beta + \omega t)$$

$When\ t = 0, q = 0, \omega\beta = n\pi$

$$q = c\sin(n\pi + \omega t) \quad where\ \ c = \sqrt{\frac{2P}{m\omega^2}}$$